Hierarchical Cancer Classification Report

AI-assisted decision support summary for breast_benign_0004.jpg

Executive Summary

Date & Time	2026-04-19 10:02:59	**Device**	cpu
Organ Model	Ready (resnet50_organ_classifier.pth)	**Subtype Model**	Ready (resnet50_subtype_classifier_best.pth)

Final Decision

Benign Breast Tissue

Outcome: ABNORMAL Confidence: 91.42%

Label: Benign Breast Tissue Pipeline Status: HIGH_CONFIDENCE

Tissue Route: Breast Tissue Subtype / Outcome Detail: Benign Breast Tissue

AI-generated decision support summary prepared for review.

Visual Review

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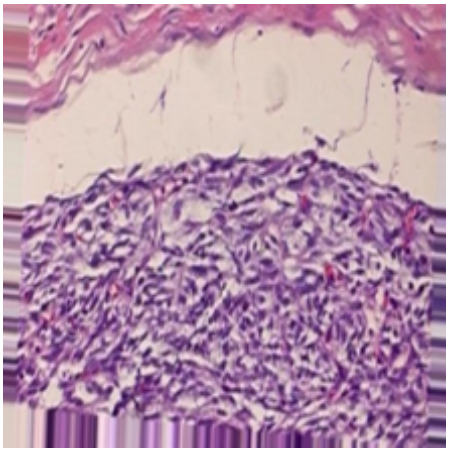
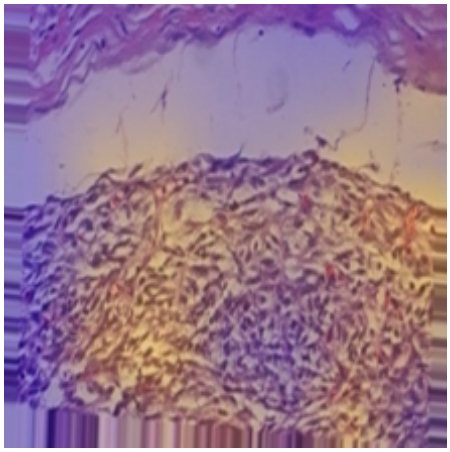
Input Scan



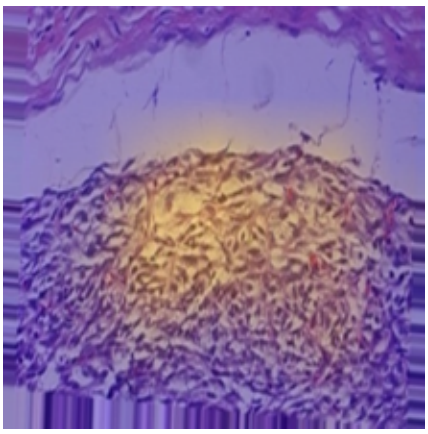
Submitted image used for analysis.

Grad-CAM: Organ Routing



Breast Tissue

Grad-CAM: Final Outcome



Benign Breast Tissue

Clinical Routing Summary

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Outcome	ABNORMAL
Confidence	91.42%
Entropy	0.29
Label	Abnormal tissue
Reason	None

Organ / Tissue Probabilities

Organ	Probability (%)
Breast Tissue	99.99%
Lymphatic System	0.01%
Brain Tissue	0.00%
Oral Tissue	0.00%
Blood (Leukemia-related)	0.00%
Lung & Colon Tissue	0.00%
Cervical Tissue	0.00%
Kidney Tissue	0.00%

Subtype Probabilities

Subtype	Probability (%)
Benign Breast Tissue	91.42%
Breast Malignancy	8.58%

Warnings / Notes

- Manual override used to continue past uncertain modality detection.
- Override used at Step 0 modality gating.